

Annexure A – Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

1. Propositional Logic Representation

Let:

- **R** = It is raining
- **W** = Ground is wet

Knowledge Base:

1. If it rains, the ground becomes wet
2. It is raining

Goal:

2. Convert into CNF

Rule

Eliminate implication:

Fact

CNF Clauses:

C1:

C2:

3. Negate the Goal

Goal:

Negated Goal:

C3:

4. Resolution Steps

Step 1

Resolve:

with

Result:

C4:

Step 2

Resolve:

with

Result:

(Empty Clause)

Conclusion

Since the Empty Clause is derived,

Therefore,

Question 2 – Student Assignment Submission**1. Propositional Logic Representation**

Let:

- **S** = Rahul submitted assignment
- **M** = Rahul receives internal marks

Knowledge Base:

Goal:

2. CNF Conversion

Rule:

becomes

Fact:

CNF Clauses:

C1:

C2:

3. Negate Goal

Goal:

Negated Goal:

C3:

4. Resolution

Step 1

Resolve

and

Result:

C4:

Step 2

Resolve

and

Result:

Conclusion

Therefore,

Question 3 – Library Membership

1. Propositional Logic

Let:

- **L** = Priya is a library member
- **B** = Priya can borrow books

Knowledge Base:

Goal:

2. CNF Conversion

Rule:

becomes

Fact:

Clauses:

C1:

C2:

3. Negated Goal

Goal:

Negated Goal:

C3:

4. Resolution

Step 1

Resolve

with

Result:

C4:

Step 2

Resolve

with

Result:

Final Conclusion

Therefore,

Question 4 – Placement Eligibility

1. Propositional Logic

Let:

- **A** = Arun cleared aptitude test
- **E** = Arun eligible for placement

Knowledge Base:

Goal:

2. CNF

Rule:

becomes

Fact:

Clauses:

C1:

C2:

3. Negated Goal

Goal:

Negated Goal:

C3:

4. Resolution

Step 1

Resolve

with

Result:

C4:

Step 2

Resolve

with

Result:

Resolution Tree

$\neg A \vee E$

|

$\neg E$

|

$\neg A$

|

A

|
□

Conclusion

Therefore,

Question 5 – Access Control System

1. Propositional Logic

Let:

- **P** = User entered correct password
- **A** = User authenticated
- **G** = User granted access

Knowledge Base:

- 1.
- 2.
- 3.

Goal:

2. Convert into CNF

Rule 1

becomes

Rule 2

becomes

Fact

Clauses:

C1:

C2:

C3:

3. Negate Goal

Goal:

Negated Goal:

C4:

4. Resolution Steps

Step 1

Resolve

with

Result:

C5:

Step 2

Resolve

with

Result:

C6:

Step 3

Resolve

with

Result:

Resolution Tree

$\neg A \vee G$

|

$\neg G$

|

$\neg A$

|

$\neg P \vee A$

|

$\neg P$

|
P
|
□

Final Conclusion

The Empty Clause is obtained.

Hence,

Therefore,

Summary

Question Goal		Result
Q1	Ground Wet	Proved
Q2	Rahul Gets Marks	Proved
Q3	Priya Can Borrow Books	Proved
Q4	Arun Eligible for Placement	Proved
Q5	User Granted Access	Proved

All five problems successfully derive the **Empty Clause (□)**, proving the goals using the **Resolution Algorithm**.